TAP: Time Series Anomaly Prediction via Adaptive Period Modeling and Dual Representation Learning

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***Abstract*—Time series anomaly detection is typically used to identify data that deviates significantly from normal data, often indicating faults or failures in the underlying system, thus facilitating system stability and safety. Most existing methods focus on detecting anomalies after they occur, while research on predicting future anomalies remains scarce. Before anomalies manifest themselves, there are often subtle precursors exhibiting slight deviations from normal behavior, with varying reaction times and intensities. Next, the setting is often characterized by a lack of labeled data, which complicates model training. To address these challenges, we propose a time series anomaly prediction framework, TAP. It can adapt flexibly to varying reaction times of anomaly precursors across different variables using a period-aware multi-scale module, and it is able to strengthen the distinction between precursors and normal sequences via a dual-branch framework that combines reconstruction and contrastive learning. The contrastive branch employs a controlled generation strategy within the multi-scale patching to produce diverse hard negative samples for precursor identification. The reconstruction branch complements this by evaluating fluctuation magnitudes to ensure sensitivity to subtle variations. We report on experiments on eight datasets from diverse domains, finding that TAP is capable of competitive or superior performance compared to baseline methods for both anomaly detection and prediction.**

***Index Terms*—Anomaly prediction, contrastive learning, reconstruction, time series.**

I. Introduction

Anomaly detection [1], [2], [3], [4], [5], [6] is a key aspect of time series analysis and is employed in scenarios such as fault identification, fraud detection, and system monitoring. Despite continuous advancements that further enhance detection robustness against noise [7] and enable precise anomaly localization [8], these detection methods fundamentally identify issues after they have occurred (Fig. 1(a)). Consequently, they still fail to meet the proactive needs for predictive maintenance.

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Anomaly prediction, a novel unsupervised time series task, aims to predict future anomalies based on current and past data (Fig. 1(b)), making it more suitable for preventive maintenance applications [9]. Studies suggest that anomalies in real-world systems often do not occur abruptly but rather evolve gradually over time [9]. In Fig. 1(b), the yellow region exhibits behaviors in the data that foreshadow the anomalies in the subsequent pink region. The behaviors that can be observed before actual anomalies occur are commonly referred to as anomaly precursors, and the duration of these behaviors is termed the reaction time. Identifying precursors offers the potential to forecast future anomalies.

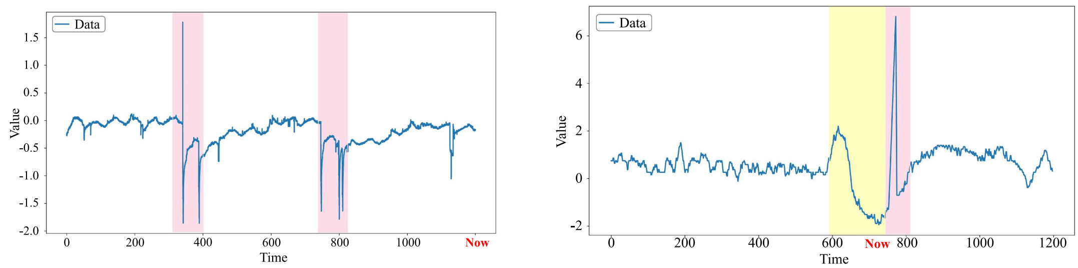
In this paper, we study the problem of time series anomaly prediction by identifying precursors of future anomalies [10]. Due to the high cost and effort of manual anomaly labeling, there has been growing interest in learning from time series data through self-supervised approaches. Although reconstruction and contrastive-learning paradigms [9], [11], [12], [13] demonstrate the effectiveness of self-supervised learning at anomaly detection, existing methods cannot be easily extended to enable anomaly prediction due to the following main challenges:

Challenge 1: Reaction times may differ across variables in multivariate time series and can also vary over time within the same variable. The focus of reconstruction-based methods on reconstructing individual data points makes it difficult to capture multiple reaction times [14], [15], [16]. Contrastive learning methods [9], [13] compare fixed-length time segments and thus struggle to model variable reaction times. However, the different variables in multivariate time series generally capture different aspects of an underlying system and can exhibit different reaction times. Existing methods fail to contend with such variable-specific reaction times.

Challenge 2: Generating negative samples that adapt to varying precursor reaction times is intrinsically difficult. Due to the absence of labeled anomalies, models must rely on negative samples to distinguish normal data from anomaly precursor data. Existing anomaly prediction methods [10] generate precursor samples by injecting perturbations at the end of a fixed-length segment. This fixed-position, fixed-length injection strategy fails to accommodate the variations in precursor locations resulting from different reaction times.

We propose a dual-branch method to time series anomaly prediction based on adaptive period modeling (**TAP**). This method is able to contend with both challenges.

To address Challenge 1, TAP introduces a period-aware multi-scale module that combines adaptive dominant period



(a) Anomaly Detection

(b) Anomaly Prediction

Fig. 1. (a) Traditional anomaly detection focuses on detecting anomalies in past time series data. The pink periods contain detected anomalies. (b) Anomaly prediction aims to identify precursor signals that exhibit slight deviations from normal data, as shown in the yellow region for the real-world PSM dataset. Such precursors often indicate the potential occurrence of future anomalies, as found in the pink region.

masking with multi-scale patching to accommodate varying reaction times. Specifically, the adaptive dominant period mask approximates the reaction time of each variable by masking its most dominant periods. Next, TAP employs multi-scale patching on the masked time series variable to extract patterns at varying granularities.

To address Challenge 2, TAP introduces a reaction-time-aware dual-branch mechanism that guides precursor injection and fluctuation modeling. Specifically, the contrastive branch injects perturbations into multi-scale patching to generate hard negative samples that simulate precursor signals. The reconstruction branch provides positive representations from the same scales and evaluates the magnitude of fluctuations under different time granularities. This dual-branch mechanism aligns perturbation injection and fluctuation modeling with estimated reaction time ranges, enabling the generation of training signals and enhancing the robustness of anomaly prediction. Our contributions are summarized as follows:

- We propose TAP, a novel multivariate time series anomaly predictor with high sensitivity to precursors and the ability to handle reaction time variations.
- TAP employs an adaptive period-aware multi-scale module that enables it to dynamically adapt to reaction time variations across variables or across time within a single variable, thereby accommodating diverse anomaly evolution patterns.
- We introduce a reaction-time aware dual-branch framework that guides precursor injection and fluctuation modeling.
- We report on extensive experiments on eight real-world time series datasets. The results show that TAP is capable of competitive or superior anomaly prediction and detection performance.

The remainder of the paper is organized as follows. Section II surveys related work, and Section III presents preliminaries and the problem definition. Section IV then presents the TAP framework, and experimental results in Section V. Section VI concludes the paper.

II. Related Work

A. Multi-Scale Learning

Inspired by the advantages of multi-scale structures in computer vision for multi-level feature extraction, multi-scale methods have been proposed for time series modeling [13], [17], [18], [19]. THOC [17] learns multi-scale representations using different skip connections for each time point in recurrent neural networks. However, its reliance on fixed skip intervals to select features limits its ability to capture gradual changes over continuous time intervals. DGHL [18] maps time series windows to a hierarchical latent space to achieve multi-scale feature learning. DCdetector [13] proposes a multi-scale attention module to reduce information loss during patching, and the final representation concatenates results from different scales. MANF [19] employs a multi-scale attention mechanism to capture local and global dependencies in time series, with the scale size increasing with the depth of the encoder layers. Neither of these earlier methods accounts for differences in fluctuation patterns across variables. Similarly, while recent advances adapt to periodic patterns for lightweight modeling [20] and channel normalization [21], they primarily focus on general forecasting. Consequently, these approaches cannot effectively capture the varying reaction times inherent in anomaly evolution.

B. Time Series Anomaly Prediction

Predicting a future anomaly based on current time series patterns is crucial in real-world settings requiring predictive maintenance, such as wind turbine or power grid settings. However, only a few studies have addressed this problem [9], [22], [23], [24], [25], most of which rely on supervised learning [22], [23], [25]. For instance, VLFEP [23] analyzes fluctuations in VLF signal amplitude and phase in the time and frequency domains and identifies noise reductions and wave excitations occurring tens of minutes before earthquakes as potential anomaly precursors. However, the reliance on precisely annotated (precursor, anomaly) pairs, which are often unavailable or expensive to obtain, makes them difficult to apply in unlabeled scenarios. To address the limitations of label dependency of supervised learning, PAD [9] proposes an unsupervised approach. Specifically, it introduces a framework based on neural-controlled differential equations (NCDEs) and adjusts thresholds for upcoming time windows that contain anomalies. Although these methods improve the ability to anticipate anomalies, they often overlook reaction time, which refers to the varying interval between the onset of a precursor and the occurrence of a full anomaly, thereby limiting their effectiveness in complex scenarios. Additionally, they model the positive samples, which prevents them from learning temporal dependencies between normal sequences and subtle precursors. IGCL [10] employs a controlled diffusion module to generate anomaly precursor patterns and predicts anomalies by identifying precursors that evolve into future anomalies. However, it injects perturbations only at the end of fixed-length segments, which makes it unable to adapt to variations in precursor locations caused by different reaction times.

C. Time Series Anomaly Detection

Time series anomaly detection has attracted notable attention in machine learning and data mining. In recent years, researchers have used a reconstruction [11], [12], [26] or a contrastive learning [9], [13] paradigm to capture data representations and patterns.

The reconstruction paradigm detects anomalies by identifying differences between the original and reconstructed time series. Early approaches, like ARIMA [27], were statistics-based. These later gave way to deep learning-based approaches [11], [12], [26], [28], [29]. OmniAnomaly [26] captures normal patterns of multivariate time series by learning robust representations of them. Anomaly Transformer [12] models prior association and series associations simultaneously to capture association discrepancies. Further, it incorporates normalized association discrepancy into its reconstruction loss, leveraging both the temporal representation and distinctiveness of association discrepancies. D^3R [11] combines decomposition and noise diffusion to reconstruct corrupted data directly. However, these methods rely either on a single scale for modeling or indiscriminately reconstruct, which prevents them from adapting to the varying reaction times of anomaly evolution.

The contrastive learning paradigm detects anomalies through discrepancies between representations derived from different aspects of the same input. DCdetector [13] employs a contrastive framework with dual attention to learn discrepancies across views, enhancing the distinction between anomalies and normal points. However, it only focuses on relationships between positive samples, which limits its performance in anomaly prediction that requires explicit negative samples. Although many contrastive paradigms [30], [31], [32], [33] have been proposed, they often focus on selecting appropriate positive samples while neglecting the impact of negative samples. However, negative samples are crucial for distinguishing anomaly precursors from normal patterns.

III. Problem Statement

We proceed to present the necessary preliminaries and then define the problem addressed. The notations are summarized in Table I.

Definition 1 (Time series): We denote a time series of length T as $\mathbf{X} = \langle \mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T \rangle \in \mathbb{R}^{T\times c}$, where $\mathbf{x}_t \in \mathbb{R}^c$ is the observation at time t and c is the dimension ality or number of variates or variables associated with an observation. When $c = 1$, $\mathbf{X}$ is univariate.

Definition 2 (Anomaly Precursor): We define an anomaly precursor as a set of signals in the time interval $[t-r, t]$, where t is the current time step. These signals begin to deviate from normal behavior and are subsequently followed by anomalies in the future interval $[t+1, t+f]$, where r and f denote the reaction time and the look-forward window [9], [23], respectively.

TABLE I
Key Notations Used in the TAP

Notation	Explanation
$\mathbf{x}_t$	Observation value of the c variable time series at time t.
$\mathbf{X}_{t-h:t}$	Historical series before time t with window length h.
r	Reaction time, which is estimated by the extracted dominant period lengths.
f	Look-forward window, representing the length of the future time interval.
$\mathbf{x}_m$	Masked time series after applying the adaptive dominant period mask.
$\mathbf{x}_{m,p}/\mathbf{x}_p$	The patch-based representations of the masked time series and the original time series at the p-th scale.
$\mathbf{A}$	Frequency amplitude obtained via FFT.
$S_{i,j}$	Frequency-domain similarity between two historical sub-sequences A_i and A_j.
$\mathbf{x}_{n,p}$	Hard negative samples generated at the p-th scale via controlled generation strategies.
z_p^+/z_p^-	Representation vectors of the positive and negative samples output by the encoder at the p-th scale.
$\hat{X}$	Reconstructed time series produced by the decoder.

Definition 3 (Anomaly Detection): Anomaly detection identifies whether anomalies exist in a given historical time series. With sliding windows, anomaly detection takes as the input historical time series $\mathbf{X}$ and outputs a binary vector $\mathbf{y} \in \mathbb{R}^T$, where $\mathbf{y}_i \in \{0, 1\}$, $i \in 1, 2, \ldots, T$, and $\mathbf{y}_i = 1$ indicates an anomaly at time i.

Definition 4 (Anomaly Prediction): Given the current real-time point t and the historical sub-sequence $\mathbf{X}_{t-h:t}$ of length h, anomaly prediction aims to output a real-time anomaly score $\hat{\mathbf{p}}_t$, which indicates the likelihood of whether an anomaly precursor exists at the current moment and an anomaly will occur in the future interval $[t+1, t+f]$ [10]. The score is converted into a binary label using a threshold μ [9], where a higher score implies a higher likelihood of a future anomaly. If $\hat{\mathbf{p}}_t \geq \mu$, the interval $[t+1, t+f]$ is considered anomalous, while it is considered normal when $\hat{\mathbf{p}}_t < \mu$.

IV. Methodology

We propose a framework, namely TAP, for anomaly prediction and detection. We first give an overview of the framework and then provide specifics on each module in the framework.

A. Overview

Fig. 2 shows the overall architecture of TAP, which consists of four main components.

The input sequence processing module normalizes the input multivariate time series using instance normalization. As variable independence is able to effectively reduce parameter complexity and mitigate overfitting [13], [34], we split the input into univariate time series. In Fig. 2, we take one univariate time series $\mathbf{x}$ as an example. All other univariate time series are modeled similarly.

The period-aware multi-scale module employs an adaptive dominant period mask, focusing on the dominant periods that reflect the underlying periodicity of each variable, thereby

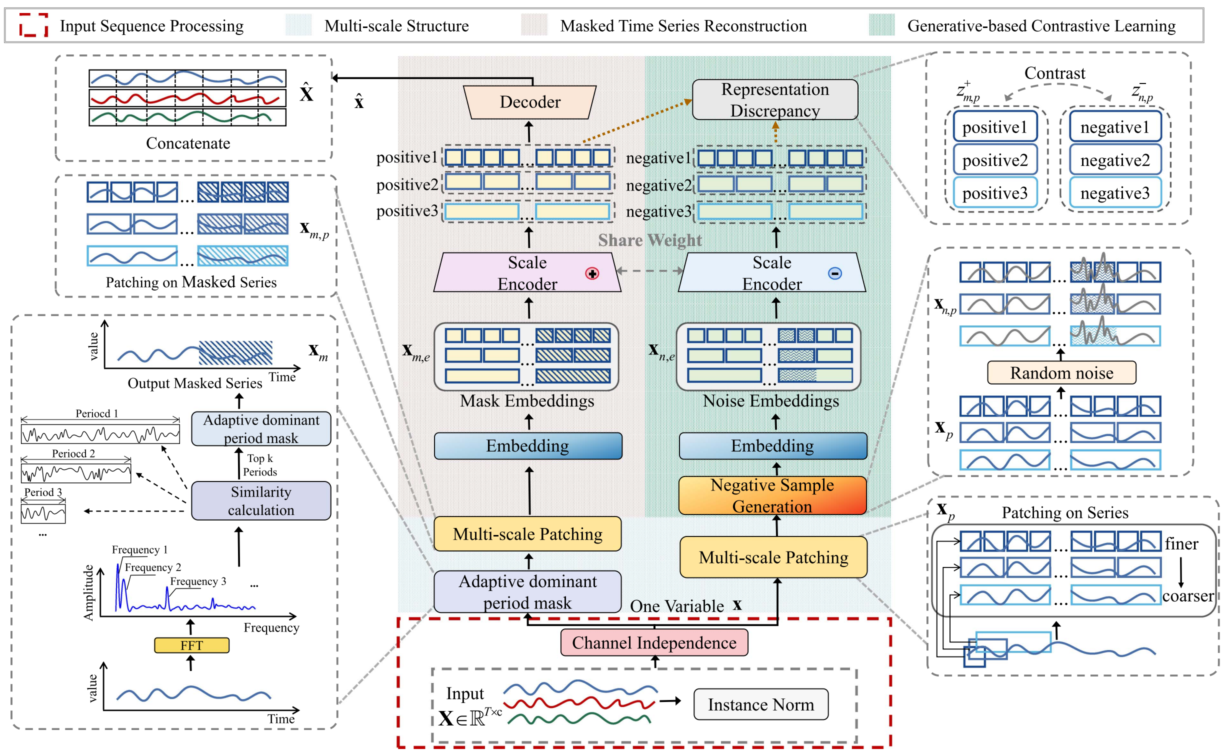


Fig. 2. Overall architecture of TAP.

distinguishing between long-term and short-term fluctuations. Additionally, multi-scale patching learns time series features at different granularities on the masked time series $\mathbf{x_m}$ and unmasked time series $\mathbf{x}$, respectively.

The *masked time series reconstruction* and *generative-based contrastive learning* modules combine to form a dual-branch structure aiming to enhance the model's sensitivity to anomaly precursors. The reconstruction branch employs an asymmetric encoder-decoder architecture that integrates multi-scale information to model fluctuation magnitudes better. The contrastive branch employs a controlled generation strategy within the multi-scale patching to produce negative samples that adapt to precursor signals under varying reaction times. The branches share the same encoder, which is composed of stacked Transformer blocks [35].

Anomaly scores are derived by combining reconstruction errors and contrastive discrepancies. Specifically, the anomaly detection score is computed on a point-wise basis, and the anomaly prediction score is derived as the average anomaly score over a historical sub-sequence.

B. Period-Aware Multi-Scale Module

Due to the diversity of scenarios and differences among variables, the reaction time of anomaly evolution varies across systems and variables. The period-aware multi-scale module captures time series features of these variations using an *adaptive dominant period mask* and *multi-scale patching*.

The adaptive dominant period mask takes as input each univariate time series $\mathbf{x}$ and produces its masked time series $\mathbf{x}_m$. Intuitively, a variable with longer periodic changes is likely to evolve gradually, while a variable with shorter periodic changes is likely to evolve more rapidly. Thus, we identify fluctuation characteristics by masking the most dominant periods for each variable.

Multi-scale patching takes as input $\mathbf{x}$ and $\mathbf{x}_m$ and produces outputs to the two branches. For the *masked time series reconstruction branch*, the masked time series $\mathbf{x}_m$ is fed to the multi-scale patching, which produces $\{\mathbf{x}_{m,p}\}_{p=1}^{s}$. For the *generative-based contrastive learning* branch, the original time series $\mathbf{x}$ is fed to the multi-scale patching, which outputs $\{\mathbf{x}_p\}_{p=1}^{s}$. Essentially, multi-scale patching enables TAP to learn time series features at various granularities, which helps the dual branch to consider varying reaction times and to differentiate between positive and negative samples.

We next introduce the adaptive dominant period mask and multi-scale patching.

Adaptive Dominant Period Mask: To capture the patterns of dominant periods, we first extract the periodic information in the frequency domain using the Fast Fourier Transform (FFT) as follows.

$$\mathbf{A} = \mathrm{Amp}\left(\mathrm{FFT}(\mathbf{x})\right), \tag{1}$$

where $\mathrm{FFT}(\cdot)$ and $\mathrm{Amp}(\cdot)$ returns the amplitude of its input so that $\mathbf{A}$ represents the amplitude of each frequency. It is known in FFT that the frequency with the largest amplitude represents the more dominant period [36].

Thus, we calculate the frequency-based similarity between the historical univariate sub-sequences before the current time t to select the most dominant periods for each variable. We use subscripts i, j to represent different historical univariate sub-sequences before the current time t, e.g, $\mathbf{A}_i = \mathrm{Amp}(\mathrm{FFT}$

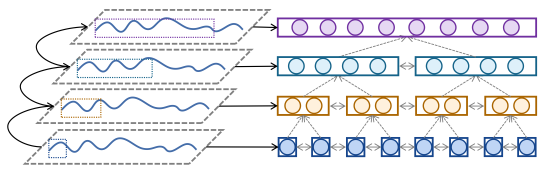


Fig. 3. Multi-scale feature extraction.

$(\mathbf{x}_{t-h:t}))$ and $\mathbf{A}_j = \mathrm{Amp}(\mathrm{FFT}(\mathbf{x}_{t-2h-1:t-h-1}))$, where h is a sliding window. We compute the frequency-based similarity as follows.

$$S_{i,j} = \frac{\mathbf{A}_i \cdot \mathbf{A}_j}{\|\mathbf{A}_i\|_2 \|\mathbf{A}_j\|_2} \tag{2}$$

Then we select the top-k frequencies by identifying the highest similarity values. This approach is justified by the fact that these frequencies reflect periodic characteristics that are highly consistent across multiple time windows within the univariate time series. If a frequency appears in multiple windows, this indicates that the frequency is dominant and consistently captures the underlying periodic pattern. Specifically, we first find the indices i_{max} and j_{max} corresponding to the maximum similarity value $S_{i,j}$. Then we select the top-k frequencies from the two most similar sub-sequences:

$$\begin{aligned} i_{max}, j_{max} &= \arg\max_{i,j}(S_{i,j}) \\ \mathbf{f} &= \arg \text{Top-}k\ (\mathbf{A}_{i_{max}}, \mathbf{A}_{j_{max}}) \end{aligned} \tag{3}$$

Specifically, as illustrated in Fig. 2, $\mathbf{f} = (f_1, \ldots, f_k) \in \mathbb{R}^k$ is the top-k dominant frequencies extracted from the historical segment up to the current time. The top-k frequencies correspond to k dominant period lengths $\mathbf{r} = (r_1, \ldots, r_k) \in \mathbb{R}^k$, where $r_k = \frac{1}{f_k}$. The dominant periods are used to estimate reaction times.

The normal data can be reconstructed well, while fluctuations are difficult to reconstruct [37]. Thus, we evaluate the magnitude of fluctuations through reconstruction errors to accurately capture anomaly precursors. We mask each univariate time series $\mathbf{x}$ with an adaptive mask length r, where r is randomly sampled from $\mathbf{r}$.

$$\mathbf{x}_m = \mathbf{M} \odot \mathbf{x}, \tag{4}$$

where $\|\mathbf{M}\|_1 = r$ and $\mathbf{M} = (0, 0, \ldots, 1, \ldots, 1) \in \{0,1\}^T$ denotes the mask matrix near the current time with adaptive mask length, and $\odot$ is element-wise multiplication. In the resulting masked time series $\mathbf{x}_m$, each variable has varying mask lengths to estimate reaction times. Building on this, we also consider the ability to more flexibly adapt to reaction times across multiple scales by including a multi-scale patching component.

Multi-scale Patching: Since masking precedes segmentation, explicit patch-level thresholding is not required. We take the masked time series $\mathbf{x}_m$ of the univariate time series $\mathbf{x}$ as an example, and we assume that three patch-based time series are constructed for $\mathbf{x}_m$, as shown in Fig. 3. These time series are segmented into multi-scale patches to accommodate varying reaction times, ranging from fine to coarse granularity, where the number of granularities is denoted by s. The finest-scale patches capture features with rapid anomaly evolution, while coarser-scale patches correspond to features generated by slower anomaly evolution. This way, TAP considers information across different scales, facilitating subsequent reconstruction strategies to enable information exchange among patches and flexibly adapt to varying reaction times.

Concretely, the multi-scale process first generates a patched time series $\mathbf{x}_1 \in \mathbb{R}^{N_1 \times P_1}$, where P_1 is the finest-granularity patch size and N_1 is the number of patches. For the i-th scale, $1 < i \leq s$, we concatenate two adjacent patches from the $(i-1)$-th scale, progressively forming coarser-scale patches, obtaining an i-th scale time series containing N_i patches with patch size P_i. By continued grouping, we obtain s time series $\{\mathbf{x}_{m,p}\}_{p=1}^{s}$ with different scales. Similarly, we obtain $\{\mathbf{x}_p\}_{p=1}^{s}$ from the unmasked time series that are used in the contrastive branch.

C. Masked Time Series Reconstruction

We perform reconstruction on the masked time series to accurately assess the magnitude of fluctuations. This module employs an asymmetric encoder-decoder architecture (see the pink region in Fig. 2). The scale encoder adopts a temporal transformer architecture, and outputs representations of positive sample pairs $\mathbf{z}_p^+ \in \mathbb{R}^{N_p \times d_{\text{model}}}$, where d_{model} denotes the hidden state dimensionality and $1 \leq p \leq s$. The decoder reconstructs the input time series using a lightweight multilayer perceptron that fuses multi-scale information, and outputs $\hat{\mathbf{X}} \in \mathbb{R}^{T \times c}$.

Scale Encoder: There are s encoder blocks, corresponding to patch time series $\{\mathbf{x}_{m,p}\}_{p=1}^{s}$ at different scales. The scale encoder is implemented based on a transformer block. First, the embedded representation $\mathbf{x}_{m,e}$ is obtained via the embedding layer from each $\mathbf{x}_{m,p}$. Then, $\mathbf{x}_{m,e}$ is fed to multi-head self-attention layers, and the representations of different patches at different scales are the units that are used to learn temporal features across different time intervals. The scale encoder provides $\{\mathbf{z}_p^+\}_{p=1}^{s}$ for both reconstruction and contrastive learning.

Reconstruction Learning: We concatenate the features $\{\mathbf{z}_p^+\}_{p=1}^{s}$ and use the multilayer perceptron-based decoder to learn cross-scale information for reconstruction. By incorporating multiple scales, we capture both fine-grained and coarse-grained temporal dependencies, allowing the model to adapt to varying reaction times. The reconstruction result can be obtained by:

$$\hat{\mathbf{X}} = \mathrm{Decoder}(\{\mathbf{z}_p^+\}_{p=1}^{s}) \tag{5}$$

TAP minimizes the mean squared error (MSE) loss. The losses across all variables are averaged to get the overall reconstruction loss:

$$\mathcal{L}_{\text{Rec}} = \|\hat{\mathbf{X}} - \mathbf{X}\|_2^2. \tag{6}$$

D. Generative-Based Contrastive Learning

We introduce controlled generative strategies to construct diverse precursor signals as negative samples for contrastive learning to better distinguish subtle anomaly precursors. The multi-scale views produced by the encoder from the masked

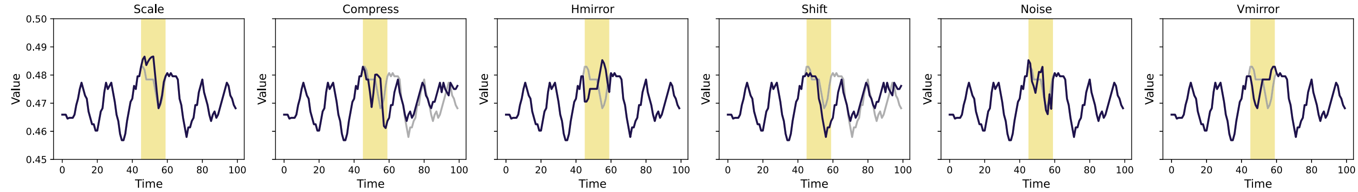


Fig. 4. Samples generation visualization.

time series reconstruction module are used as positive samples for contrastive learning, denoted as $\{\mathbf{z}_p^+\}_{p=1}^s$. In anomaly prediction, the representation of positive samples $\{\mathbf{z}_p^+\}_{p=1}^s$ is denoted as $\mathbf{z}_{p[i]}^{pre}$, where i is the variable index. For anomaly detection, the representation of the positive samples $\{\mathbf{z}_p^+\}_{p=1}^s$ is represented as $\mathbf{z}_{p[q]}^{det}$, where q ranges over time points.

Negative sample generation: For multi-scale patch time series $\mathbf{x}_p$, we apply noise pollution techniques to generate hard negative samples $\mathbf{x}_{n,p}$. Specifically, *scale* adjusts the amplitude by a random factor (magnifying or reducing) to simulate precursors characterized by gradual amplification or attenuation, such as slow sensor drift. *Compress* alters the time resolution of the original series to simulate scenarios where the evolution process suddenly accelerates. *Hmirror* flips the series around its mean value (horizontal axis) to simulate structural precursors where the polarity of local fluctuations is inverted (e.g., an expected upward spike becomes a downward dip). *Shift* displaces the time series forward to modify temporal alignment without changing amplitude, which simulates precursors exhibiting an advanced onset relative to the expected periodic structure. *Noise* introduces stochastic Gaussian perturbations to simulate random small fluctuations. Finally, *vmirror* reverses the time series along the chronological axis to simulate precursor patterns with a distorted temporal evolution order. A visualization is shown in Fig. 4.

Taking Gaussian noises as examples, each original data point from different scales is randomly added with Gaussian noise, causing negative samples to vary in different scales at different points [38]. The negative data after noise addition is:

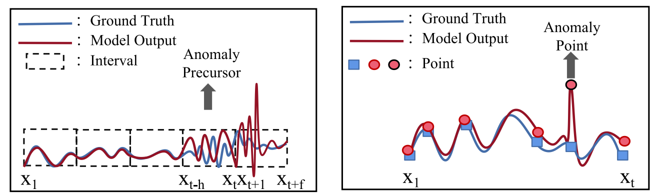
$$\mathbf{x}_{n,p} = \mathbf{x}_p \cdot (\sigma \cdot s_p + 1), \tag{7}$$

where s_p represents the random Gaussian noises for scale p, and σ is the noise intensity randomly sampled from a uniform distribution $[0, \sigma_{max}]$. The random Gaussian noises are sampled from a standard Gaussian white noise distribution:

$$s_p \sim \mathcal{N}(0, 1) \tag{8}$$

We sample s_p independently across scales to ensure distinct random factors for each scale, enabling diverse negative samples and allowing the model to focus on normal features at different scales from the positive samples $\{\mathbf{x}_p\}_{p=1}^s$.

The embedded representation $\mathbf{x}_{n,e}$ is obtained via the embedding layer from the generated negative samples $\mathbf{x}_{n,p}$. The encoder backbone is shared with the masked time series reconstruction module. Specifically, the generated data is fed into the backbone network to derive the hard negative sample representations $\mathbf{z}_p^- \in \mathbb{R}^{N_p \times d_{\text{model}}}$, where $1 \leq p \leq s$. For anomaly



(a) Prediction (b) Detection

Fig. 5. Task-specific loss functions. (a) Prediction: Reaction time deviations denote precursors for future window $\mathbf{x}_{t+1:t+f}$. (b) Detection: High loss signifies anomalies.

prediction, the representation is $\mathbf{z}_{neg(p)}^{pre}$, and for anomaly detection, it is $\mathbf{z}_{neg(p)}^{det}$.

Representation discrepancy: As shown in Fig. 5, we use different loss functions for different tasks. For anomaly prediction, we use an interval-wise contrastive loss function to compute the differences between positive and negative samples (Fig. 5(a)). Three views with different patching granularities are used as an example. Since anomaly prediction focuses on detecting anomalies in specific segments of the time series, we apply mean pooling to the various views generated by the scale encoder to obtain interval-level representations for each variable. We use the interval representations from the same variable i as positive pairs $\mathbf{z}_{1[i]}^{pre}$, $\mathbf{z}_{2[i]}^{pre}$, $\mathbf{z}_{3[i]}^{pre}$. Similarly, we have three hard negative samples $\mathbf{z}_{neg(1)}^{pre}, \mathbf{z}_{neg(2)}^{pre}, \mathbf{z}_{neg(3)}^{pre}$. The representations from other variables are also taken as negative samples. The loss calculation is $\mathcal{L}_{\text{con}}^{inter}$ defined as follows.

$$\mathcal{L}_{\text{con}}^{inter}(\mathbf{z}_1^{pre}) = -\frac{1}{\mathcal{C}} \sum_{i=1}^{c} log\left(\frac{\mathcal{Z}_{1,2} + \mathcal{Z}_{1,3}}{\mathcal{Z}_{1,k} + \mathcal{Z}_{1,neg1}}\right), \tag{9}$$

where $\mathcal{Z}_{1,2} = \exp(\mathbf{z}_{1[i]}^{pre} \cdot \mathbf{z}_{2[i]}^{pre})$, $\mathcal{Z}_{1,3} = \exp(\mathbf{z}_{1[i]}^{pre} \cdot \mathbf{z}_{3[i]}^{pre})$, $\mathcal{Z}_{1,k} = \sum_{i \neq j} \sum_{k=1}^{3} \exp\ (\mathbf{z}_{1[i]}^{pre} \cdot \mathbf{z}_{k[j]}^{pre})$, $\mathcal{Z}_{1,neg(1)} = \sum_i \exp\ (\mathbf{z}_{1[i]}^{pre} \cdot \mathbf{z}_{neg(1)}^{pre})$. $\exp(\mathbf{z}_{1[i]}^{pre} \cdot \mathbf{z}_{2[i]}^{pre})$ represents the inner product of the interval representations of $\mathbf{z}_1^{pre}$ and $\mathbf{z}_2^{pre}$.

Intuitively, normal samples should remain close to each other across different scales, as they share similar underlying characteristics. In contrast, anomalous samples are difficult to be close to. Therefore, this design can capture the consistency of normal samples, thus strengthening the sensitivity of the model to small deviations. The overall contrastive objectives are defined as follows.

$$\mathcal{L}_{Con} = \frac{1}{s} \sum_{p=1}^{s} \mathcal{L}_{\text{con}}^{inter}(\mathbf{z}_p^{pre}) \tag{10}$$

For the anomaly detection task, we design a point-wise contrastive loss function (Fig. 5(b)). To temporally align patch-level representations with the original resolution without introducing interpolation artifacts, we apply a deterministic upsampling operation:

$$\mathbf{z}^{det} = Upsampling(\mathbf{z}_{\xi(q,l_{\text{orig}},l_{\text{new}})})$$
$$\xi(q, l_{\text{orig}}, l_{\text{new}}) = \left\lfloor \frac{q \cdot l_{\text{orig}}}{l_{\text{new}}} \right\rfloor, \tag{11}$$

where ξ is the mapped index, $\lfloor . \rfloor$ is the floor function, $q \in \{0, 1, \ldots, l_{\text{new}} - 1\}$ is the upsampled time index, z is either $\mathbf{z}_p^+$ or $\mathbf{z}_p^-$, l_{orig} is the original patch length, and l_{new} is the target window size. By assigning existing patch representations to time indices rather than interpolating new values.

Representations of different views at the same time point q are treated as positive pairs $\mathbf{z}_{1[q]}^{det}$, $\mathbf{z}_{2[q]}^{pre}$, $\mathbf{z}_{3[q]}^{pre}$. Negative pairs include: representations between different time points within the same view, the representation of any time point in $\mathbf{z}_1^{det}$ compared with all time points in $\mathbf{z}_2^{det}$ and $\mathbf{z}_3^{det}$ except that point, the representation of any time point in the hard negative sample $\mathbf{z}_{neg(1)}^{det}$, $\mathbf{z}_{neg(2)}^{det}$ and $\mathbf{z}_{neg(3)}^{det}$. The loss calculation $\mathcal{L}_{\text{con}}^{point}$ defined as follows.

$$\mathcal{L}_{\text{con}}^{point}(\mathbf{z}_1^{det}) = -\frac{1}{T}\sum_{i=1}^{T} log\left(\frac{\mathcal{Z}_{1,2}^{det} + \mathcal{Z}_{1,3}^{det}}{\mathcal{Z}_{1,k}^{det} + \mathcal{Z}_{1,neg1}^{det}}\right), \tag{12}$$

where $\mathcal{Z}_{1,2}^{det} = \exp(\mathbf{z}_{1[q]}^{det} \cdot \mathbf{z}_{2[q]}^{det})$, $\mathcal{Z}_{1,3}^{det} = \exp(\mathbf{z}_{1[q]}^{det} \cdot \mathbf{z}_{3[q]}^{det})$, $\mathcal{Z}_{1,k}^{det} = \sum_{u \neq q} \sum_{k=1}^{3} \exp(\mathbf{z}_{1[q]}^{det} \cdot \mathbf{z}_{k[u]}^{det})$ and $\mathcal{Z}_{1,neg1}^{det} = \sum_i \exp(\mathbf{z}_{1[q]}^{det} \cdot \mathbf{z}_{neg(1)}^{det})$.

In summary, the different designs of the contrastive loss enable the same fundamental algorithm to be applied flexibly according to different task requirements, enhancing the practicality of the model.

E. Joint Optimization

As previously mentioned, reconstruction and contrastive learning are interconnected. The reconstruction loss focuses on extracting the magnitude of anomalies that deviate from the normal. The contrastive loss effectively learns whether the normal deviates in terms of overall trends and patterns. The overall loss function is given as follows.

$$\mathcal{L} = \lambda_{\text{Con}}\mathcal{L}_{\text{Con}} + \lambda_{\text{Rec}}\mathcal{L}_{\text{Rec}}, \tag{13}$$

where λ_{Con} and λ_{Rec} control the contributions of the two losses. Since the deviations at the anomaly evolution stage are relatively subtle, a larger weight is assigned to the reconstruction loss to accurately evaluate the magnitude of deviations. Unless stated otherwise, we set $\lambda_{\text{Rec}} = 0.7$ and $\lambda_{\text{Con}} = 0.3$.

F. Model Inference

During inferencing, a labeled negative sample construction is not needed. The anomaly score is composed of the MSE between the input and the reconstructed output, as well as the representational distance between positive sample pairs, which can be determined using metrics such as Euclidean distance.

TABLE II
DETAILS OF ORIGINAL DATASETS

Datasets	Domain	Train	Valid	Test	Dim	AR (%)
MSL	Spacecraft	46,653	11,664	73,729	55	10.5
SMAP	Spacecraft	108,146	27,037	427,617	25	12.8
SWAT	Water treatment	396,000	99,000	449,919	51	5.78
NIPS-TS-SWAN	Space Weather	48,000	12,000	60,000	38	32.6
NIPS-TS-GECCO	Water treatment	55,408	13,852	69,261	9	1.1
SMD	Server Machine	566,724	141,681	708,420	38	4.2
PSM	Server Machine	105,984	26,497	87,841	25	27.8
UCR	Various Natural Sources	1,790,679	447,670	6,143,541	1	0.6

For anomaly detection, the final anomaly score $\mathbf{y}_i$ is defined as follows.

$$\mathbf{y}_i = MSE(\hat{\mathbf{X}}, \mathbf{X}) + Dist(\mathbf{z}_p^+, \mathbf{z}_j^+)_{p \neq j, p, j=1,..,a}, \tag{14}$$

which is a point-wise anomaly score. By applying a threshold [11], we can determine whether a point is abnormal. For anomaly prediction, an anomaly score $\hat{\mathbf{p}}_t$ is produced to indicate the presence of current anomaly precursors and the likelihood of future anomalies. The final anomaly score is computed as the average over the historical sub-sequence.

V. Experiments

A. Experimental Setup

Datasets: We evaluate TAP on eight real-world datasets: (1) MSL (Mars Science Laboratory rover) [39] includes operational data from multiple sensors on the Mars rover. (2) SMAP (Soil Moisture Active Passive satellite) [39] provides soil moisture information collected from satellite sensors. (3) SMD (Server Machine Dataset) [26] is a large-scale dataset collected over five weeks from a large Internet company. (4) PSM (Pooled Server Metrics) [40] comprises data collected from eBay's application server nodes. (5) SWaT (Secure Water Treatment) [41] is a dataset for security research in water treatment systems. (6) NIPS-TS-SWAN [42] is a multivariate time series dataset extracted from solar photospheric vector magnetograms. (7) NIPS-TS-GECCO [43] covers data collected from sensors in a drinking water supply system. (8) UCR [44] contains 250 sub-datasets from various sources. Additional information on the datasets can be found in Table II. The "Train," "Valid," and "Test" columns indicate the number of samples in the training, validation, and test sets, respectively. The "Dim" column indicates the dimensionality of multivariate datasets, and the "AR(%)" column denotes the proportion of anomalies.

Baselines: We compare our method against PAD [9] and IGCL [10], both of which are unsupervised anomaly prediction approaches. Additionally, we follow the PAD approach to adapt several state-of-the-art anomaly detection methods for anomaly prediction, thereby obtaining ten baselines. These include the reconstruction-based methods: CAE-Ensemble [37], GANomaly [45], OmniAnomaly [26], Anomaly Transformer (A.T.) [12], D^{3}R [11]; contrastive learning-based methods: DCdetector [13], PAD [9]; the classic methods: IForest [46]; a density-estimation method: DAGMM [1]; a clustering-based method: Deep SVDD [47].

PAD [9] presents a neural controlled differential equation-based neural network to enable both anomaly detection and

Precursor-of-Anomaly (PoA) detection. IGCL [10] uses a diffusion module to generate fixed-position precursor patterns for anomaly prediction. CAE-Ensemble [37] proposes a CNN-based autoencoder and diversity-driven ensemble learning. GANomaly [45] adopts an Encoder1-Decoder-Encoder2 architecture and introduces the idea of adversary-training to achieve its unsupervised learning scheme. OmniAnomaly [26] extends the LSTM-VAE model, using reconstruction probabilities for detection. LSTM [48] proposes forecasting-based approaches that forecast values and, based on the forecasting errors, detect whether time points are anomalous. Anomaly Transformer [12] focuses on the relationship between adjacent points. D^3R[11] decomposes unstable data into stable and trend components, and then reconstructs the data directly after it has been corrupted by noise. DCdetector [13] proposes a contrastive learning-based dual-branch attention architecture without considering reconstruction errors. IForest [46] employs the degree of distance between points and regions to find abnormal points. DAGMM [1] is a deep automatic coding Gaussian mixture model that comprises a compression network and an estimation network. Deep SVDD [47] finds an optimal hypersphere in a feature space learned using a neural network. Points that are far from the center of the sphere are considered anomalies.

Evaluation Criteria: Recent studies [49], [50] provide evidence that the point adjustment strategy used in some previous studies [12], [13], [51], [52] is problematic. This strategy treats an entire anomalous segment as correctly detected if even a single point in it is correctly identified, which may lead to inflated performance findings and may mask the actual detection accuracy of a method. Therefore, we do not adopt the point adjustment strategy in our experiments. For anomaly prediction, we follow the evaluation protocol used in existing studies [9], [10] and report the three standard classification metrics: Precision (P), Recall (R), and F1-score (F1) [26]. For anomaly detection, we employ the affiliation-based metrics that have recently become widely used in detection tasks [11], [50], including: Affiliation Precision (Aff-P), Affiliation Recall (Aff-R), and Affiliation F1-score (Aff-F1). As both precision and recall are highly sensitive to the choice of threshold, focusing on either metric alone may lead to biased conclusions. Therefore, following many existing studies [12], [13], [53], we emphasize the use of the F1 and Aff-F1 scores as balanced indicators of model performance. We also employed the AUC-ROC (AUC) metric [54] for both tasks.

Experimental Settings: We have implemented all baselines using the official or open-source versions available on GitHub and have carefully tuned their hyperparameters based on the recommendations from the original papers. For the anomaly detection methods, we follow the evaluation protocol proposed in PAD [9], and we adapt them for the anomaly prediction task. The target label from the look-forward window is modified for the anomaly prediction task following PAD [9]. In the experiments, TAP is trained using the Adam optimizer [49] with an initial learning rate of 1e-5. The model contains three encoder layers. The multi-scale patch sizes are 2, 4, and 8. For anomaly prediction, the default look-forward window f is 4. Early stopping with a patience of 3 epochs is employed against the validation loss during training. Following existing studies [11], [26], [55], we choose the best threshold for all methods. All experiments are conducted using Python 3.9 and PyTorch 1.13 [56] on CUDA 12.0, with an NVIDIA Tesla-A800 GPU.

B. Performance Evaluation

Table III presents an anomaly prediction performance comparison of the models, in terms of precision, recall, and F1-score. On all five real-world datasets, TAP achieves the best F1 performance.

Anomaly prediction: We first compare TAP with the two representative anomaly prediction baselines, PAD and IGCL. TAP consistently achieves significant improvements over IGCL across multiple datasets, with F1-score gains of 3.26% on MSL, 4.68% on SMAP, 2.57% on SMD, 5.59% on PSM, and 3.57% on SWaT. This performance gap mainly stems from TAP's reaction time-aware dual-branch design, which generates negative samples at varying temporal ranges and aligns them with reconstruction-based fluctuation modeling. In contrast, IGCL injects perturbations only at the end of fixed-length segments, rendering it unable to adapt to variable precursor positions resulting from diverse reaction times.

PAD performs considerably worse than both TAP and IGCL. Although PAD is the first to introduce unsupervised anomaly prediction by leveraging NCDE-based modeling, this approach fails to explicitly model precursor–normal dependencies, which are critical for robust anomaly prediction, and it does not explicitly handle reaction time variability. These limitations significantly restrict its effectiveness.

Among the baseline models considered, traditional machine learning methods like IForest exhibit inferior performance compared to the deep learning-based methods. This is because the traditional methods do not account for the sequential nature and underlying complexities of time series data, resulting in limitations in generalization. Although DCdetector leverages contrastive learning on dual-channel representations within and across patches, its effectiveness is limited. This can be attributed to its reliance solely on positive samples, which limits its capacity to effectively learn anomaly precursors, thereby limiting its performance at anomaly prediction that requires label support from negative samples. Compared to the previous state-of-the-art model, D^3R, TAP achieves notably improved F1 scores on the benchmark datasets.

Furthermore, the performance on the PSM and SWaT datasets is markedly better than on other datasets, likely because anomaly prediction requires the model to detect anomaly precursors. In some datasets, precursors may be very sparse, which limits performance.

As shown in Table IV, we evaluate the performance of TAP on the NIPS-TS-SWAN and NIPS-TS-GECCO datasets using ACC, P, R, F1, and AUC [57]. These datasets are more challenging due to their wide variety of anomaly types and varying anomaly ratios, with NIPS-TS-SWAN having one of the highest anomaly rates (32.6%) and NIPS-TS-GECCO one of the lowest (1.1%). Even in these more complex anomaly

TABLE III
ANOMALY PREDICTION RESULTS FOR FIVE REAL-WORLD DATASETS. SUPERIOR PERFORMANCE IS INDICATED BY HIGHER METRIC VALUES, WITH THE TOP F1 SCORES EMPHASIZED IN BOLD AND THE SECOND-BEST UNDERLINED

	MSL			SMAP			SMD			PSM			SWaT			Average
Method	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1	F1
DAGMM	13.94	17.04	15.33	11.91	22.00	15.51	14.74	15.37	15.05	39.27	36.14	37.64	82.14	61.72	70.48	30.80
Iforst	13.82	17.11	15.29	11.79	18.67	14.45	15.25	18.28	16.63	39.39	38.82	39.10	75.39	62.22	68.17	30.72
Deep SVDD	12.94	16.04	14.32	11.77	17.09	13.94	14.99	15.78	15.37	36.63	36.30	36.46	81.29	61.95	70.31	30.08
LSTM	16.18	15.28	15.72	11.68	22.65	15.41	14.21	16.42	15.24	39.20	40.31	40.02	66.76	68.36	67.55	30.78
A.T.	14.62	12.42	13.43	11.40	22.80	15.20	10.52	19.42	13.65	30.56	34.64	32.47	75.00	60.50	66.97	28.34
DCdetector	2.35	3.46	2.80	8.04	11.30	9.40	3.71	9.03	5.26	28.97	12.05	17.02	46.45	36.36	40.79	15.05
PAD	14.37	13.22	13.77	15.19	34.41	21.08	11.23	18.76	14.05	31.23	33.05	32.11	74.83	59.09	66.04	27.21
Omni	13.66	21.66	16.75	11.90	23.39	15.77	16.22	18.19	17.15	39.45	40.88	40.15	85.73	58.57	69.59	31.88
GANomaly	15.36	17.30	16.27	12.18	23.41	16.02	15.06	21.39	17.68	37.02	43.06	39.81	83.99	59.77	69.84	31.92
CAE-Ensemble	20.35	18.27	19.26	15.88	26.68	19.91	18.41	19.51	18.94	40.18	41.99	41.07	88.61	59.90	71.48	34.13
D^{3}R	19.99	20.32	20.15	15.73	26.89	19.85	15.78	19.33	17.38	40.73	42.21	41.46	84.13	61.85	71.29	34.02
IGCL	20.43	21.69	21.04	17.09	27.09	20.96	16.44	25.65	20.04	41.22	44.99	43.02	84.21	63.19	72.20	35.45
TAP	16.27	41.66	**23.41**	16.59	47.11	**24.53**	19.99	23.27	**21.51**	36.77	65.30	**47.05**	99.38	60.29	**75.05**	**38.49**

TABLE IV
ANOMALY PREDICTION RESULTS FOR THE NIPS-TS DATASETS. ALL RESULTS ARE IN %

Dataset	Method	ACC	P	R	F1	AUC
NIPS-TS-SWAN	A.T.	58.19	52.20	38.75	44.48	52.98
	DCdetector	74.83	21.04	10.94	14.40	50.58
	CAE-Ensemble	58.57	55.98	41.52	47.68	64.81
	D^{3}R	58.38	55.32	42.64	48.16	63.87
	IGCL	60.12	57.01	42.75	48.86	64.63
	TAP	61.75	71.28	55.22	**62.23**	66.82
NIPS-TS-GECCO	A.T.	89.41	43.62	27.92	34.05	50.39
	DCdetector	97.21	1.76	3.46	2.33	50.79
	CAE-Ensemble	97.71	51.67	29.92	37.90	52.94
	D^{3}R	98.01	51.24	30.91	38.56	53.09
	IGCL	98.42	52.10	32.77	40.23	53.11
	TAP	99.10	82.89	27.39	**41.17**	53.35

TABLE V
OVERALL PREDICTION RESULTS ON UNIVARIATE DATASETS

Dataset	Method	ACC	P	R	F1	AUC
UCR	A.T.	87.04	32.07	20.86	25.28	51.68
	DCdetector	87.80	12.59	10.90	11.68	48.65
	CAE-Ensemble	88.46	34.15	21.47	26.37	58.26
	D^{3}R	90.82	39.10	20.77	27.13	60.11
	IGCL	91.52	35.63	23.57	27.74	64.45
	TAP	94.12	20.64	50.74	**29.34**	76.02

environments, TAP consistently achieves strong performance, suggesting applicability across different domains.

We evaluate univariate time series by training/testing each sub-dataset separately, and Table V shows that TAP achieves the best average performance.

TAP incorporates a period-aware multi-scale module with adaptive dominant period masking to capture periodic variations and anomaly precursors across different time horizons. To further address the challenge of generating negative samples that adapt to varying precursor timings, TAP adopts a reaction time-aware dual-branch framework. The contrastive branch injects perturbations within the multi-scale patching process to generate hard negative samples that simulate precursor signals at different onset positions and reaction times. In parallel, the reconstruction branch provides positive representations from the same scales and evaluates fluctuation magnitudes across multiple temporal resolutions. Sharing a common Transformer-based encoder, the two branches jointly align precursor injection with fluctuation modeling, thereby producing high-quality training signals. These designs overcome the limitations of fixed-position strategies in previous work and maintain robust performance across diverse benchmark datasets.

Anomaly detection: As shown in Table VI, TAP also achieves the best overall performance at anomaly detection. It performs 1.51%-5.97% higher on average than the previous SOTA methods. As explained already, we disregard the point adjustment strategy and use other metrics for evaluation. Metrics based on affiliation offer more objective and reasonable assessments, resulting in lower scores for existing methods.

Table VII presents a performance comparison between TAP and other well-performing methods on the NIPS-TS-SWAN and NIPS-TS-GECCO datasets. Although the two datasets have the highest (23.68% in NIPS-TS-SWAN) and lowest (1.25% in NIPS-TS-GECCO) anomaly ratios, TAP demonstrates consistently good performance across these challenging datasets. On the NIPS-TS-SWAN dataset, TAP improves on the best baseline model by 7.11% in terms of the key metric Aff-F1 (from 43.19 to 50.30). On NIPS-TS-GECCO, it shows an improvement of 4.02% (from 78.68 to 82.70).

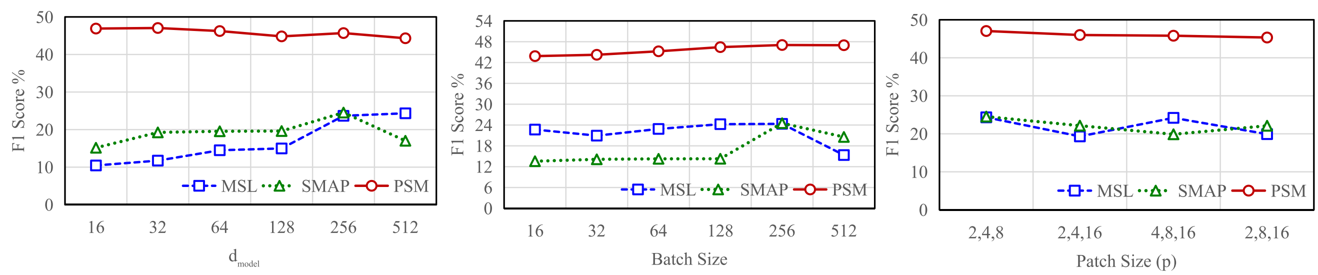
C. Ablation Studies

We proceed to conduct ablation studies focusing on key components of TAP: the period-aware multi-scale module, the masked time series reconstruction, and the generative-based contrastive learning. The experiments aim to provide a comprehensive evaluation of the method while assessing the effectiveness of its key components. The details of these variants are outlined below:

- *w/o multi-scale:* This variant removes the multi-scale patching, using only a single-scale time series.
- *w/o adaptive mask:* Replaces the adaptively adjusted mask scale with a fully random mask.
- *w/o reconstruction:* This variant removes the masked reconstruction loss.
- *w/o contrastive:* This variant removes the contrastive learning loss.
- *w/o generated samples:* Eliminates the noise generation.

TABLE VI
ANOMALY DETECTION ON FIVE TIME-SERIES DATASETS. THE BEST AFF-F1 SCORES APPEAR IN BOLD, AND THE SECOND-BEST ARE UNDERLINED

	MSL			SMAP			SMD			PSM			SWaT			Average
Method	Aff-P	Aff-R	Aff-F1	Aff-P	Aff-R	Aff-F1	Aff-P	Aff-R	Aff-F1	Aff-P	Aff-R	Aff-F1	Aff-P	Aff-R	Aff-F1	Aff-F1
DAGMM	4.07	92.11	68.14	50.75	96.38	66.49	63.57	70.83	67.00	68.22	70.50	69.34	59.42	92.36	72.32	68.65
IForest	53.87	94.58	68.65	41.12	68.91	51.51	71.94	94.27	<u>81.61</u>	69.66	88.79	78.07	53.03	99.95	69.30	69.82
Deep SVDD	49.88	98.87	65.73	42.67	68.23	51.94	65.84	80.43	72.58	58.32	60.11	59.95	55.73	97.34	70.77	64.19
LSTM	58.82	14.68	23.49	55.25	27.70	36.90	60.12	84.77	70.35	57.06	95.92	71.55	49.99	82.11	62.15	52.88
A.T.	51.04	95.36	66.49	56.91	96.69	71.65	54.08	97.07	66.42	54.26	82.18	65.37	53.63	98.27	69.39	67.86
DCdetector	55.94	95.53	70.56	53.12	98.37	68.99	50.93	95.57	66.45	54.72	86.36	66.99	53.25	98.12	69.03	68.40
PAD	56.33	82.21	68.15	41.67	64.52	53.94	59.54	67.66	63.71	68.45	57.72	59.21	54.73	92.35	68.06	62.61
Omni	51.23	99.40	67.61	52.74	98.51	68.70	79.09	75.77	77.40	69.20	80.79	74.55	62.76	82.82	71.41	71.93
GANomaly	56.36	98.27	68.01	56.44	97.62	72.52	73.46	83.25	74.20	55.31	98.34	75.11	59.93	81.52	70.24	72.01
CAE-Ensemble	54.99	93.93	69.37	62.32	64.72	63.50	73.05	83.61	77.97	73.17	73.66	73.42	62.10	82.90	71.01	71.05
D^3R	66.85	90.83	<u>77.02</u>	61.76	92.55	<u>74.09</u>	64.87	97.93	78.02	73.32	88.71	<u>80.29</u>	60.14	97.57	<u>74.39</u>	<u>76.76</u>
TAP	67.94	93.03	**78.53**	66.66	89.94	**76.57**	75.36	94.85	**83.99**	73.36	93.13	**82.08**	62.52	97.26	**76.12**	**79.45**



(a) d_{model} (b) Batch size (c) Multi-scale patch size

Fig. 6. Parameter sensitivity studies of main hyperparameters for anomaly prediction.

TABLE VII
EXPERIMENTAL RESULTS FOR ANOMALY DETECTION ON THE NIPS-TS DATASETS. ALL NUMBERS ARE PERCENTAGES

Dataset	Method	ACC	Aff-P	Aff-R	Aff-F1	AUC
NIPS-TS-SWAN	A.T.	63.89	48.12	25.89	33.67	44.74
	DCdetector	66.54	44.47	8.61	14.42	43.46
	D^3R	68.83	68.73	31.49	<u>43.19</u>	37.31
	TAP	67.84	63.28	41.74	**50.30**	59.48
NIPS-TS-GECCO	A.T.	95.82	50.39	88.68	64.27	51.60
	DCdetector	97.50	52.07	90.79	66.18	45.38
	D^3R	98.90	77.83	79.56	<u>78.68</u>	80.72
	TAP	97.67	71.68	97.71	**82.70**	93.11

We also conducted ablation experiments for anomaly detection, and Table VIII shows the ablation study results for anomaly detection.

Effectiveness of the Multi-Scale Framework: The results show that performance decreased across all datasets after removing the multi-scale framework. Notably, some datasets experience a clear drop in performance, which is due to their dependence on the multi-scale framework caused by varying reaction time lengths.

Impact of Masking Strategy: Using a random mask instead of the designed frequency-based mask and periodicity-based mask also reduces performance. By considering the varying temporal evolution speeds among different variables, the ability of TAP to adaptively capture the reaction differences of each variable is enhanced.

Role of Masked Reconstruction and Contrastive Learning: The experiments reveal that removing either the masked reconstruction module or the contrastive learning component leads to a considerable drop in performance. This is due to the resulting reduced ability to effectively identify fluctuations and magnitude changes, which are essential for anomaly prediction. The absence of these components prevents the model from fully leveraging the complementary nature of reconstruction and contrastive learning, resulting in suboptimal performance across all datasets.

Effectiveness of Sample Generation Strategy: The sample generation strategy contributes to improvements in performance, with an increase of 2.85% (from 20.56 to 23.41). The generated samples enhanced the ability of the model to distinguish between normal patterns and precursors.

In the ablation studies for anomaly detection, we define the ablation models as for anomaly prediction. The results in Table IX show the negative effects of removing components on the overall performance.

D. Discussion and Analysis

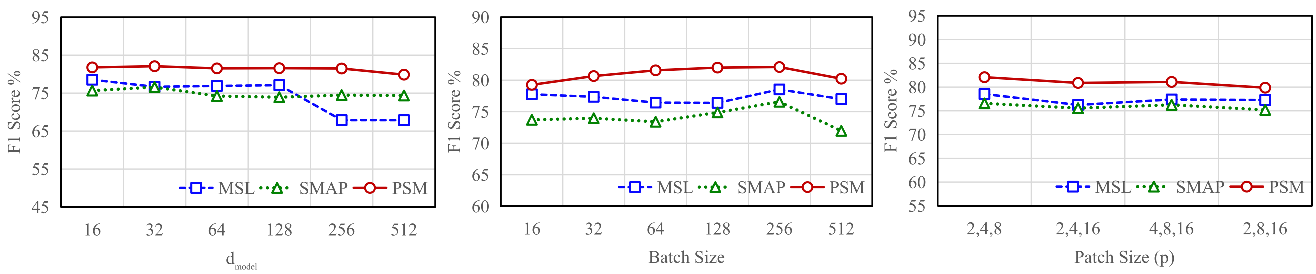
Hyperparameter analysis: The hyperparameters that might influence the performance of TAP include the hidden state dimensionality, the batch size, and the multi-scale patch size. To assess their impact on performance, we conduct a hyperparameter sensitivity analysis on the MSL, SMAP, and PSM datasets. Fig. 6 and 7 show the effects of varying the hyperparameters for anomaly prediction and detection, respectively. Specifically, Fig. 6(a) reports the performance when varying the size of the latent space, as the performance of many deep neural networks is influenced by d_{model}. Fig. 6(b) shows the effects of training TAP with different batch sizes. Fig. 6(c) presents the

TABLE VIII
ABLATION RESULTS FOR ANOMALY PREDICTION. THE BEST F1 SCORES ARE HIGHLIGHTED IN BOLD

	TAP			w/o multi-scale			w/o adaptive mask			w/o reconstruction			w/o contrastive			w/o generation		
Dataset	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1
MSL	16.27	41.66	**23.41**	11.50	53.89	18.96	15.20	47.76	23.06	7.69	34.19	12.56	11.62	51.85	18.99	12.62	55.52	20.56
SMAP	16.59	47.11	**24.53**	15.48	31.76	20.82	14.94	39.95	21.75	14.94	26.92	19.22	16.63	40.41	23.56	16.37	40.05	23.24
PSM	36.77	65.30	**47.05**	42.39	50.27	46.00	35.84	62.00	45.42	28.54	35.60	31.68	36.33	65.17	46.65	37.03	63.79	46.86

TABLE IX
ABLATION RESULTS FOR ANOMALY DETECTION. THE BEST F1 SCORES ARE HIGHLIGHTED IN BOLD

	TAP			w/o multi-scale			w/o adaptive mask			w/o reconstruction			w/o contrastive			w/o generation		
Dataset	Aff-P	Aff-R	Aff-F1	Aff-P	Aff-R	Aff-F1	Aff-P	Aff-R	Aff-F1	Aff-P	Aff-R	Aff-F1	Aff-P	Aff-R	Aff-F1	Aff-P	Aff-R	Aff-F1
MSL	67.94	93.03	**78.53**	65.87	93.57	77.31	67.48	90.32	77.25	44.65	44.09	44.37	67.01	93.08	77.92	67.37	93.09	78.17
SMAP	66.66	89.94	**76.57**	64.28	90.34	75.11	66.41	88.31	75.81	53.74	76.61	63.17	65.39	90.07	75.77	65.65	86.16	74.52
PSM	73.36	93.13	**82.08**	76.15	80.59	78.31	72.29	90.38	80.33	52.94	69.90	60.25	73.22	92.43	81.71	73.53	92.74	82.03



(a) d_{model} (b) Batch size (c) Multi-scale patch size

Fig. 7. Parameter sensitivity studies of main hyperparameters for anomaly detection.

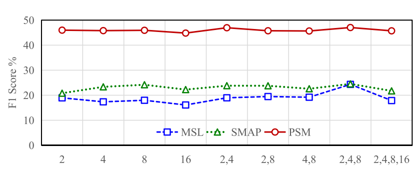


Fig. 8. Anomaly prediction at different scales (F1).

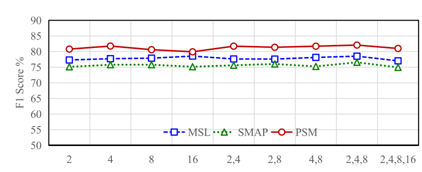


Fig. 9. Anomaly detection at different scales (Aff-F1).

performance of TAP across varying multi-scale configurations. For some datasets, the performance is relatively stable across different patch size combinations. However, slight fluctuations are observed on other datasets, probably due to more pronounced differences in dominant frequencies in the data. To balance the information capture ability between scales, the default patch sizes are uniformly set to 2, 4, and 8.

Scale Analysis: Multi-scale approaches have a unique impact on anomaly prediction, particularly when faced with varying reaction times. The prediction and detection performances at different scales are shown in Fig. 8 and 9, respectively. First, a single scale is only suitable for a specific range of reaction times and struggles to adapt to the diverse range of reaction times in the data. Second, different datasets exhibit varying dependencies on the patch sizes. Using the PSM dataset as an example, when the patch combination is set to 2, 4, the performance surpasses that of the 2, 8 and 4, 8 combinations. This indicates that the dataset is characterized more heavily by shorter time scales.

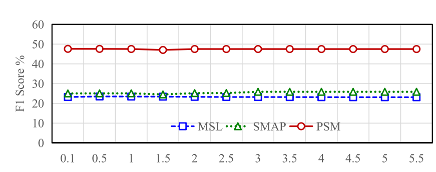


Fig. 10. F1 scores for different max noise intensity.

Moreover, having more scales is not necessarily better. When the gap between the smallest and largest scales becomes too large, the difficulty of the contrastive learning process increases, potentially leading to inconsistencies in representations across scales, which reduces performance.

Noise intensity: As shown in Fig. 10, the performance across different datasets remains relatively stable when varying the max noise intensity σ_{max}. This indicates that TAP is robust to changes in noise levels and maintains its ability to distinguish

TABLE X
COMPARISON OF TRAINING RUNTIME (ONE EPOCH)

Dataset	MSL	SWaT	SMAP
Method	Runtime(s)	Runtime(s)	Runtime(s)
PAD	248	2376	244
GANomaly	81	993	77
CAE-Ensemble	62	787	59
D^3R	133	1443	128
TAP	49	459	37
w/o multi-scale	42	449	32
w/o adaptive mask	47	378	34
w/o reconstruction	46	455	36
w/o contrastive	18	439	29
w/o generated samples	37	442	30

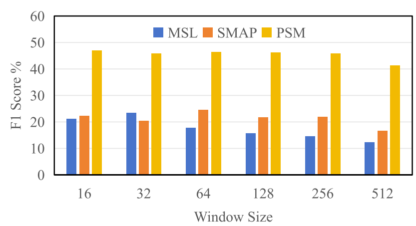


Fig. 11. Performance under different sliding window sizes.

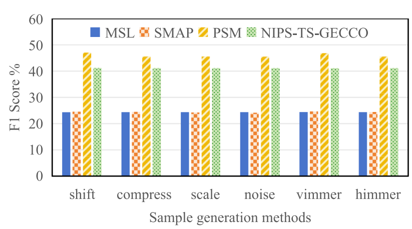


Fig. 12. F1 scores for six sample generation methods.

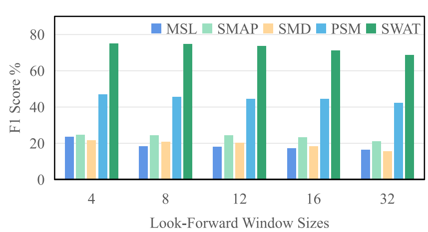


Fig. 13. F1 scores for different look-forward window sizes.

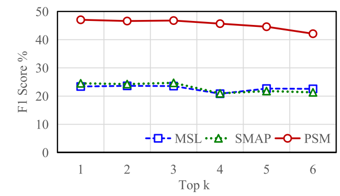


Fig. 14. Results with different values.

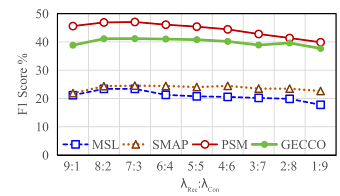


Fig. 15. Results with different combinations.

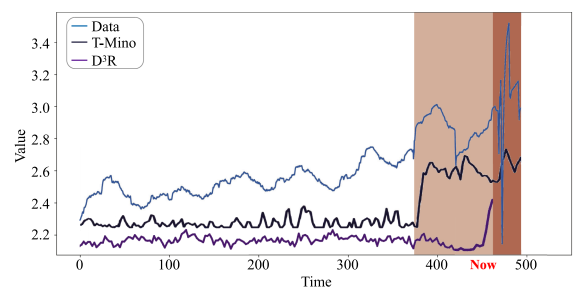


Fig. 16. Visualization of anomaly prediction on the PSM dataset. The blue line represents the actual time series, while the black line indicates the output of TAP and the purple line represents the output of D^3R. TAP assigns larger scores to anomaly precursors (pink region) that exhibit early deviations before the occurrence of severe anomalies (red region).

subtle anomaly precursors from normal patterns under different perturbation intensities.

Time Window Analysis: We conduct an analysis of window size performance on three datasets without multi-scale support, as shown in Fig. 11. This study follows existing methodologies [48], [58], adjusting the sliding window size. The results indicate that the best performance on the PSM dataset is obtained with a relatively short window size of 16, suggesting that its fluctuations change quickly. In contrast, the best performance on the MSL and SMAP datasets is obtained with larger window sizes of 32 and 64, respectively, reflecting that their fluctuations evolve over a relatively longer period. This finding highlights the importance of multi-scale modeling. Without it, we would need to adjust the window size to accommodate the varying time series features.

Sample generation: The anomaly prediction results for different generation methods are shown in Fig. 12. While TAP performs well across all methods, temporal shifting slightly outperforms Gaussian noise. Gaussian noise introduces point-wise fluctuations that create easily distinguishable negative samples. Conversely, temporal shifting preserves amplitude and local shape, generating harder negative samples that compel the contrastive branch to learn deeper representations for identifying subtle precursors.

Look-forward windows: To analyze the impact of the prediction distance, we conduct experiments with different look-forward window sizes. The results are shown in Fig. 13. As the prediction distance increases, the difficulty of prediction also increases gradually, causing a decline in performance. This phenomenon is in line with the expectation that when the look-forward window is smaller, meaning the prediction distance is shorter, the fluctuations in reaction time are more closely related to future subsequences. However, as the look-forward window expands, the model needs to consider data changes and potential dynamic patterns over a longer period, requiring stronger generalization capabilities and the ability to capture long-term dependencies.

Top-k selection: In Fig. 14, we analyze the sensitivity to k and find that stable performance is achieved with small values (e.g., $k \in 1, 2, 3$), while larger k introduces noisy periodic components and degrades performance. Hence, k is set to a small value.

Loss balance: In Fig. 15, we perform a sensitivity analysis over different settings of λ_{Rec} and λ_{Con}. The results show that $(0.7, 0.3)$ yields the best and most stable performance. This setting remains effective even under extremely low anomaly ratios, e.g., on NIPS-TS-GECCO.$\lambda_{\text{Rec}} : \lambda_{\text{Con}}$

E. Running Time Analysis

In Table X, we present the overall runtime of different state-of-the-art methods and of TAP after removing key components. TAP runs faster than the baselines. TAP leverages multi-scale parallel processing and feature extraction, enabling it to focus on relevant information at different granularities, effectively reducing the processing of irrelevant features. These results demonstrate that our model design ensures efficiency while achieving high accuracy.

F. Visual Analysis

We visualize the anomaly prediction results of TAP on the PSM dataset in Fig. 16. Here, TAP identifies the onset of anomaly precursors at timestamp 375 (the pink region), preceding the occurrence of severe anomalies after timestamp 460 (the red region).

VI. Conclusion

We propose TAP, a multi-scale framework for anomaly prediction and detection on multivariate time series. The framework encompasses reconstructive and contrastive branches. To address anomaly reaction time variations across variables, TAP introduces an adaptive dominant-period masking mechanism to approximate the different reaction times of variables, enabling it to capture reaction-time fluctuations. A multi-scale patching strategy is incorporated to extract features at varying granularities, enhancing TAP's ability to handle anomalies evolving at different speeds. To improve sensitivity to subtle anomaly precursors, TAP integrates support for the capture of existence and magnitude characteristics of precursors into the framework. TAP's contrastive branch leverages multi-scale patching across different temporal ranges and focuses on distinguishing subtle precursors from normal patterns through the controlled generation of hard negatives. The reconstructive branch evaluates the magnitude of these deviations via multi-scale reconstruction. Extensive experiments on eight real-world datasets show that TAP is capable of consistent, competitive, or superior anomaly prediction and detection performance. Overall, TAP offers a proactive and adaptive approach to time series anomaly analysis, which is important in settings where early warnings and predictive maintenance are critical.

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